THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **Patterns in Life: Diversity and Classification, Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** classification	taxonomy
2	**(c)** Animalia	heterotrophic multicellular
3	**(c)** chitin	absorptive nutrition
4	**(a)** air quality	colour changes with pollutants
5	**(d)** vascular tissue	dependent on moisture
6	**(a)** yeast and hydra	outgrowth
7	**(a)** cutting	artificial vegetative propagation
8	**(c)** penicillin	from Penicillium
9	**(c)** meiosis	halves the chromosome number
10	**(d)** ovary	each has an egg cell

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Simple body with rhizoids; no vascular tissue; need water for fertilisation (1+1).	1+1
12	Plants with true roots, stems, leaves and vascular tissue but no seeds (1); fern (1).	1+1
13	So that the zygote restores the full number after fertilisation (2).	2
14	Sepals, petals, stamens and pistil (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	They have needle-like leaves that reduce water loss (1.5); they do not need water for fertilisation and form seeds (1.5).	1½+1½
16	Producing improved varieties with desired characters (1); selective breeding and artificial hybridisation (2).	1+2
17	Porifera: pores, no tissues, e.g. sponge (1.5); Cnidaria: tissue level, tentacles, one opening, e.g. hydra (1.5).	1½+1½
	OR: Flatworms: flat, bilateral, no body cavity; roundworms: cylindrical, with a body cavity and complete gut (3).	3
18	Many eggs are destroyed by water currents or eaten by other animals (1.5); more eggs increase the chances of survival (1.5).	1½+1½
	OR: External: outside, fish and frogs, many eggs; internal: inside the body, birds and mammals, fewer eggs and more protection (1.5+1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Fern (1). (ii) Moss (1). (iii) Yes (1). (iv) Fern, as vascular tissue transports water and supports a larger body (2).	1+1+1+2
	OR: (i) Seeds naked vs enclosed; no flowers vs flowers (2). (ii) Monocots and dicots (1). (iii) Wheat, maize; pea, mango (2).	2+1+2
20	(i) Site, number of eggs, examples (2). (ii) The embryo is protected inside (1). (iii) Fertilisation in a dish, then the embryo placed in the uterus (2).	2+1+2
	OR: (i) Testes, sperm duct, urethra; ovaries, oviducts, uterus, vagina (3). (ii) Testes: sperms and hormones; ovaries: eggs and hormones (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **Patterns in Life: Diversity and Classification, Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** dicots	e.g. pea, mustard
2	**(c)** Cnidaria	stinging cells
3	**(a)** molluscs	soft bodies
4	**(b)** mammary glands and hair	feed milk to young
5	**(c)** Kingdom, Phylum, Class, Order, Family, Genus, Species	broad to specific
6	**(c)** fruit	ovule becomes the seed
7	**(c)** reptiles, birds and mammals	inside the female body
8	**(c)** uterus	womb
9	**(a)** father (X or Y)	mother gives X always
10	**(a)** HIV/AIDS	use of condoms prevents it

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gymnosperms have naked seeds with no fruit; angiosperms have seeds enclosed in fruits (1+1).	1+1
12	Monocots have one cotyledon; dicots have two (1+1).	1+1
13	Transfer of pollen from anther to stigma (1); wind, water, insects, birds (1).	1+1
14	The ovary becomes the fruit (1); the ovule becomes the seed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Segmented body; organ system level; muscles for movement; e.g. earthworm (1+1+1).	1+1+1
16	Egg → larva → pupa → adult for butterfly (2); the larva feeds and the adult emerges with a different body (1).	2+1
17	Hard exoskeleton, jointed appendages, segmented body, adaptation to land, water and air (3).	3
	OR: Molluscs: soft body with a shell, e.g. snail; echinoderms: spiny skin, internal skeleton of calcium carbonate, e.g. starfish (1.5+1.5).	1½+1½
18	A pair of ovaries produce eggs (1); oviducts carry the egg (1); the uterus holds the developing foetus and opens into the vagina (1).	1+1+1
	OR: Testes produce sperms (1.5); sperms travel through the sperm duct and urethra (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Porifera, Cnidaria, Platyhelminthes, Nematoda, Annelida, Arthropoda, Mollusca, Echinodermata (4). (ii) Cnidaria (1).	4+1
	OR: (i) Annelida and Arthropoda (1). (ii) Beetle (1). (iii) Protection, muscle support and survival on dry land (1). (iv) Arthropoda and Annelida (2).	1+1+1+2
20	(i) Puberty (1). (ii) Hormones (1). (iii) Onset of menstruation, development of breasts (1). (iv) Supports rapid growth (2).	1+1+1+2
	OR: (i) Ovulation and thickening of the lining and menstruation (2). (ii) The zygote implants in the uterus and pregnancy starts (2). (iii) It varies from 21 to 35 days among women (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **Patterns in Life: Diversity and Classification, Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the name given by local people	scientific basis
2	**(c)** Monera	prokaryotes
3	**(a)** unicellular fungus	chitin wall
4	**(c)** thallophytes (algae)	no differentiation
5	**(b)** pteridophytes	ferns
6	**(d)** bread mould (Rhizopus)	light spores in air
7	**(d)** Louis Pasteur	germ theory
8	**(c)** gametes from two parents combine	mixing of chromosomes
9	**(c)** stamen	anther + filament
10	**(c)** pollination	wind, water, insects

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A flexible rod-shaped structure (1); chordates (1).	1+1
12	(i) Cnidaria (1) (ii) Annelida (1).	1+1
13	External: outside the body in water; internal: inside the female body (1+1).	1+1
14	Sperm (male) and egg or ovum (female) (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Primitive chordates with a notochord at some stage of life (2); Amphioxus (1).	2+1
16	Egg matures in the ovary and is released (1); the uterine lining thickens (1); if there is no fertilisation, the lining sheds as menstruation (1).	1+1+1
17	Fishes (rohu), amphibians (frog), reptiles (lizard), birds (crow), mammals (human) (3).	3
	OR: Birds: feathers, wings, four-chambered heart; mammals: hair, mammary glands, live young (1.5+1.5).	1½+1½
18	Add 28 days to 5 March (1.5); 2 April (1.5).	1½+1½
	OR: Mother's eggs always carry X (1); father's sperm carries X or Y (1); XX gives a girl, XY a boy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Fishes, amphibians, reptiles, birds, mammals (3). (ii) Gills and fins, feathers and hollow bones, fur and fat (2).	3+2
	OR: (i) Two-part naming by Linnaeus (1). (ii) Genus first with a capital; species small; italics or underlined (2). (iii) Panthera tigris; Mangifera indica (2).	1+2+2
20	(i) Embryo, foetus growth, readiness for birth (3). (ii) Nourishment, oxygen and waste removal (1). (iii) Smoking and alcohol (1).	3+1+1
	OR: (i) Balanced diet rich in proteins, vitamins and minerals (1). (ii) Complete nutrition and protection from infection (1). (iii) Anxiety or low mood after delivery that can be treated (1). (iv) To prevent infections in mother and baby (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET D

Chapters 11 & 12: **Patterns in Life: Diversity and Classification, Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** monocots	one cotyledon
2	**(a)** flatworms	Platyhelminthes
3	**(c)** Echinodermata	spiny skin
4	**(b)** Amphibia	live in water and on land
5	**(b)** species	can interbreed and produce fertile offspring
6	**(c)** seed	contains the embryo
7	**(b)** sperms	produced in the testes
8	**(a)** ovulation	mid-cycle
9	**(d)** placenta	umbilical cord
10	**(c)** IVF	test-tube baby

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Jointed legs and a hard exoskeleton (1+1).	1+1
12	Two-part naming system by Linnaeus (1); Homo sapiens (1).	1+1
13	Release of an egg from the ovary (1); the thick uterine lining is shed as menstruation (1).	1+1
14	Provides oxygen and nutrients to the foetus and removes its wastes (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Kingdom → Phylum → Class → Order → Family → Genus → Species (2); each level has more shared features, e.g. for humans, Animalia, Chordata, Mammalia, Primates (1).	2+1
16	First trimester: embryo forms and organs begin; second: foetus grows and movement is felt; third: rapid growth and preparation for birth (1+1+1).	1+1+1
17	Local names differ and cause confusion (1); genus name capital and species name small; italics or underlined (2).	1+2
	OR: It has many endemic species under threat (1.5); sacred groves are traditionally protected forest patches that conserve local species (1.5).	1½+1½
18	Condoms (barrier), oral pills (hormones), copper-T (IUD), surgical methods (3).	3
	OR: Prevents infections (1); wash regularly, change pads in 4–6 hours, dispose of pads properly (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Kingdom to species, each narrower (2). (ii) Animalia, Chordata, Mammalia, Primates, Hominidae, Homo, sapiens (2). (iii) Members interbreed and produce fertile offspring (1).	2+2+1
	OR: (i) A structure that reduces damage, like a wall (1). (ii) Mangroves in Odisha (1). (iii) They break down pollutants (1). (iv) Forests protect people, soil, water and species (2).	1+1+1+2
20	(i) Prevents unplanned pregnancy and infections (1). (ii) HIV/AIDS, herpes; use of condoms and avoiding contact with infected persons (2). (iii) Condoms, pills, copper-T with how each works (2).	1+2+2
	OR: (i) It forms gametes with half the chromosome number (1). (ii) 23, 23 and 46 (2). (iii) Each gamete receives a random combination of chromosomes from each pair (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.